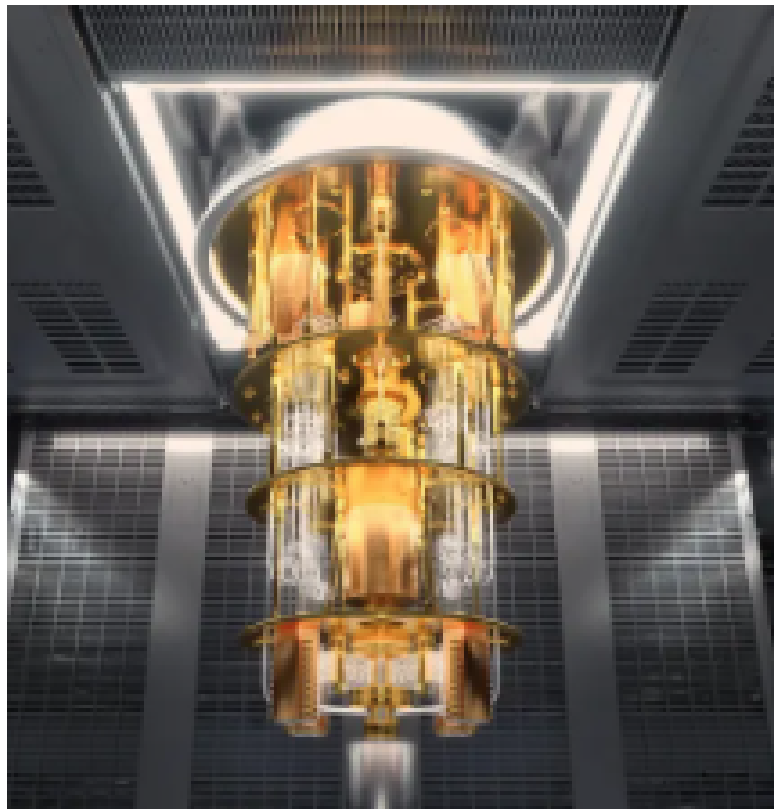


Decision Theory and Expert Judgment
TU Delft

A Structured Expert Judgment Study on the Quantum Computing Boom

Forecasting the next major technology wave



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Abstract

This report presents a Structured Expert Judgment (SEJ) study on the future of quantum computing, carried out with the Classical Model (CM) of Roger Cooke. While quantum computing is frequently described as the next major technology revolution, the timing and magnitude of this “quantum boom” remain highly unpredictable. Because there is almost no historical data to build a purely data-driven model, this is precisely the setting for which SEJ was designed. A panel of eight experts gave 5%, 50% and 95% quantiles for fourteen calibration questions with known answers and five questions of interest about the future of the field. To test if the domain background changes the conclusions, the panel was analyzed in three different configurations: the full panel of eight, a sub-panel of four assessors with a quantum-physics background, and a sub-panel of four with an investment / private-equity background. In all three configurations, the performance-weighted decision maker outperformed the equal-weight one on the combined calibration-information score, and the strongest decision maker overall was the one built from the full panel (calibration 0.399, information 1.37), which concentrates its weight on the only two well-calibrated assessors, one from each background group. The aggregate forecast projects that the global quantum-computing market will exceed USD 50 billion around **2039** (90% interval 2032-2050) and the first cryptographically relevant break of RSA-2048 around **2049** (90% interval 2035-2067). The two background groups disagree on the timing of the “Q-day”¹. This divergence may indicate a degree of overconfidence among domain experts.

¹The term “Q-Day” (or Quantum Day) refers to the hypothetical milestone when a quantum computer becomes powerful enough to break modern public-key encryption

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1 Introduction

1.1 Motivation

Big technology waves tend to follow a recognizable pattern: first a foundational technology matures, then investment surges, and finally massive adoption follows, often with a speculative overvaluation and a subsequent market correction. Three historical examples make this cycle clear.

- **The World Wide Web (1993–2002).** Tim Berners-Lee invented the Web in 1991. By 1995 there were already around 16 million internet users worldwide [7], a number that grew more than fivefold by 2000. The dot-com boom that accompanied this adoption peaked in March 2000, when the NASDAQ Composite reached its all-time high; the crash that followed erased more than five trillion dollars of market value by 2002 (Figure 1).
- **Mobile and smartphones (2007–2015).** The launch of the iPhone in 2007 caused a massive smartphone adoption, app ecosystems and a generation of mobile-first services, reshaping entire industries within a decade.
- **Artificial intelligence (2012–present).** The deep-learning revolution started with AlexNet (2012), a convolutional neural network architecture developed for image classification tasks. By 2022 generative tools such as ChatGPT had reached mass public awareness, with private investment in generative AI growing roughly tenfold in two years [12].

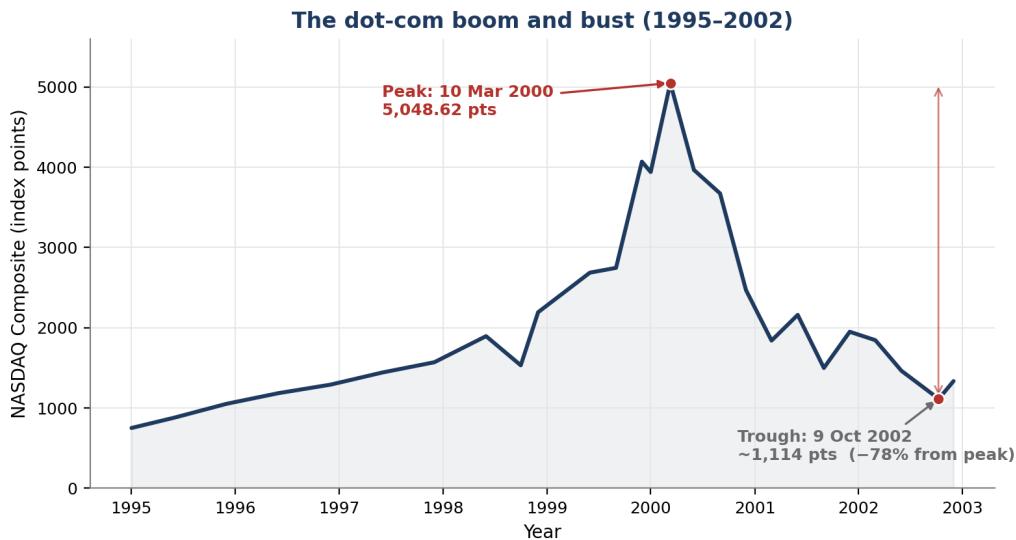


Figure 1: The dot-com boom and bust. Reconstruction of the NASDAQ Composite from documented anchor points: the index closed at its all-time peak of 5,048.62 points on 10 March 2000 and closed at near 1,114 points on 9 October 2002, a decline of roughly 78% [10]. The shape between these points is illustrative.

Each of these waves shares the same three-act structure, first a foundational breakthrough, then a surge of capital and attention, and finally broad adoption, and in each case the early uncertainty about *when* and *how big* was enormous. The question this report addresses is whether quantum computing is now entering the same kind of wave, and if so, on what timeline.

1.2 The current state of quantum-computing

Quantum computing exploits quantum-mechanical phenomena such as superposition, entanglement and interference, and it has the potential to deliver exponential speed-ups in cryptography, drug discovery, materials simulation and financial optimisation. By early 2025, governments worldwide had committed roughly USD 10 billion to quantum research, and about 24 companies were offering more than 40 quantum processing units (QPUs) commercially [9]. The commercial market itself was estimated at only USD 1–3 billion in 2024, and estimates of the number of operational machines worldwide are contested and depend heavily on definitions (gate-based, annealing, cloud-accessible).

To reason about orders of magnitude it helps to know the figures of the previous waves. Table 1 collects the reference quantities used throughout this study; they serve as sanity checks rather than as forecasts.

Table 1: Reference quantities from previous technologies waves used in the elicitation, drawn from the sources cited in the questionnaire.

Quantity	Value
Internet users worldwide, 1995	~16 million
arXiv submissions, all subjects, 2023	~208,000
quant-ph submissions, 2016	~4,500
AI-related publications worldwide, 2013	~102,000
AI patents filed globally, 2010	~3,800
Total global VC, all sectors, 2021 (peak)	~USD 620 bn
GenAI private investment, 2022	~USD 3.4 bn
Public investment in quantum research by early 2025	~USD 10 bn
Commercial QPUs offered (24 vendors), 2025	>40
Active non-quantum supercomputers, 2025	~9,000

1.3 Why Structured Expert Judgment

Despite the above data, the timing and magnitude of a possible quantum boom remain deeply uncertain. There is no long time series of revenue, adoption or performance to extrapolate from, and the few numbers that exist are contested. Standard forecasting tools struggle in this regime of *deep uncertainty* because there is no stationary data-generating process to fit and extrapolation of recent growth rates can be wildly misleading near an inflection point. However, there exist people who understand the technology and the market. Structured Expert Judgment turns their opinions into quantified probability distributions. The Classical Model of Cooke [4] is the most established SEJ method because it does not simply average opinions, it weights each expert by their demonstrated performance on questions whose answers are known (calibration questions), and it is built to satisfy the principles of rational consensus, which are reproducibility, accountability, empirical control, neutrality and fairness.

1.4 Contribution and structure of the report

Apart from running the Classical Model on a single panel, this study also aims to answer another question: *does the background of the assessors change what we conclude about the future of quantum computing?* To answer it we build and compare three decision makers, one from the full panel and one from each of the two background sub-panels (quantum-physics vs. investment).

This report will begin with the theory behind the Classical Model, which can be found in Section 2. Section 3 describes the elicitation protocol, the panel and the three-panel design, and presents the calibration and interest questions in detail. Section 4 reports expert and decision-maker performance for the three configurations, including robustness checks. Section 5 presents the aggregate forecasts for the questions of interest, Section 6 discusses the divergence between the two background groups and the limitations, and Section 7 concludes. Two appendices give the full decision-maker matrices and the questionnaire reference material.

2 The Classical Model

The Classical Model treats expert opinion as a source of scientific data. A central premise of the model is that some experts provide more informative and reliable probabilistic judgments than others. Consequently, each expert is assigned a weight based on their performance on a set of calibration questions for which the true values are known. These weights reflect the accuracy and statistical quality of the experts' judgments. The model is designed to satisfy the principles of rational consensus that an expert judgment study should satisfy: reproducibility, accountability, empirical control, neutrality and fairness [4].

These principles deserve a brief comment since they form the foundation upon which the Classical Model is constructed. First, *reproducibility* requires that the data and the procedure used to analyze them be fully documented, so that a peer reviewer can repeat the calculations and obtain the same result. Next, *accountability* means that the identity of the participating experts in the study should be available for scrutiny. *Empirical control* is the property that gives the method its scientific character. The assessments are tested against quantities whose true values are known, so that the quality of an expert is measured rather than assumed. *Neutrality* requires that the scoring and combination scheme reward honest reporting and remain robust to bias, whether intentional or not, which is precisely what the strictly proper scoring rule introduced below guarantees. Finally, *fairness* demands that no expert should be favored or penalized before the assessments are actually scored, so that all participants enter the analysis on equal conditions. All of these principles explain why the Classical Model relies on calibration questions and on performance-based weights, rather than on reputation, credentials or on the experts' self-assessment.

The two scores that determine these weights, together with the procedure that aggregates the experts into a single decision maker, are explained in the rest of this chapter.

2.1 Expert weights

Each expert is asked to provide quantiles for several uncertain quantities. As standard in the Classical Model, we elicit the 5%, 50% and 95% quantiles. The weight of an expert e combines a *calibration score* $C(e)$, which measures statistical accuracy, and an *information score* $I(e)$, which measures how informative (how narrow) the assessments are. The non-normalized weight is

$$w_e = C(e) \cdot I(e) \cdot \mathbf{1}_{\{C(e) \geq \alpha\}}, \quad (1)$$

where the indicator $\mathbf{1}_{\{C(e) \geq \alpha\}}$ gives weight zero to any expert whose calibration score falls below a threshold α . The weights are normalized to sum to one,

$$\tilde{w}_e = \frac{w_e}{\sum_{k=1}^E w_k}, \quad (2)$$

with E being the number of experts. Notice that the score is a *strictly proper scoring rule* asymptotically, namely, an expert maximizes their expected weight if and only if the stated assessments match their true beliefs. This approach eliminates incentives for strategic reporting, as experts cannot improve their influence by artificially widening or narrowing their uncertainty intervals. In practice, calibration dominates the product: experts who are poorly calibrated receive negligible weights, even when their assessments are highly informative.

2.2 Calibration score

The calibration score evaluates how accurately an expert's assessed uncertainty intervals capture the true realizations of the calibration variables. Eliciting the 5%, 50% and 95% quantiles divides

the real line into $n = 4$ inter-quantile bins with theoretical probability vector

$$p = (p_1, p_2, p_3, p_4) = (0.05, 0.45, 0.45, 0.05). \quad (3)$$

For an expert with m calibration questions we record the bin into which each realization falls and form the empirical vector $s = (s_1, s_2, s_3, s_4)$, where s_i is the fraction of realizations in bin i . A perfectly calibrated expert reproduces p . The discrepancy between s and p is the relative information or Kullback-Leibler divergence

$$I(s, p) = \sum_{i=1}^n s_i \ln \left(\frac{s_i}{p_i} \right). \quad (4)$$

Under the null hypothesis that the realizations are independent draws from p , the statistic $2m I(s, p)$ is asymptotically χ^2 with $n - 1$ degrees of freedom. The asymptotic nature of this statistic, and the influence of the number of seed variables on the reliability of the resulting scores, are examined by [2] and [5]. We return to this point in Section 6.5 when discussing the moderate number of seeds we used here. The calibration score is the p -value of this test,

$$C(e) = 1 - \chi_{n-1}^2(2m I(s, p)), \quad (5)$$

so $C(e) \in (0, 1)$, with values close to 1 indicating excellent calibration and values near 0 indicating that the realizations are very unlikely under the declared distributions (typically overconfidence). In our study $m = 14$; the quality of the χ^2 approximation for such a moderate number of calibration questions is examined critically in Section 6.

2.3 Information score

The information score measures how concentrated an expert's distribution is with respect to a background measure chosen by the analyst. A high score corresponds to a confident expert with narrow intervals, and a low score to an uncertain expert with wide intervals.

Before the score can be computed, a distribution must be reconstructed from the three elicited percentiles. The Classical Model uses the *minimally informative* distribution consistent with them; that is, that the probability mass is spread uniformly inside each inter-quantile bin, so that the four bins receive the masses 0.05, 0.45, 0.45, 0.05. This construction needs a bounded support. When a natural support exists it is used directly. In the case of describing a percentage, for instance, this natural interval is $[0, 100]$. Otherwise the support is estimated from the assessments themselves. Letting $i = 1, \dots, N$ be the indexes of the experts and writing $(q_5^i, q_{50}^i, q_{95}^i)$ for their percentiles on a given question, we define

$$L := \min_{1 \leq i \leq N} \{q_5^i, \text{realization}\}, \quad U := \max_{1 \leq i \leq N} \{q_{95}^i, \text{realization}\}, \quad (6)$$

where the realization is included for calibration questions and ignored for the questions of interest. The support is then the *intrinsic range*

$$[L_*, U_*] = [L - k(U - L), U + k(U - L)], \quad (7)$$

where k is the overshoot, an analyst-defined hyperparameter conventionally set to $k = 0.1$. On this support the analyst places a low-information background measure P , with the aim of avoiding the introduction of substantive prior knowledge. In practice, either a uniform or a log-uniform measure is used, however, the second one is preferred when the percentiles span several orders of magnitude. Both the support and the background measure may differ from question to question.

With the distribution and the background measure fixed, the information score of expert i on question j is the relative information of the expert's density f_i^j with respect to the background density p^j ,

$$I_i^j = \int_S f_i^j(x) \ln \left(\frac{f_i^j(x)}{p^j(x)} \right) dx. \quad (8)$$

Since no realization enters this definition, the information score can be computed for every question, including the questions of interest. For a uniform background measure and four inter-quantile bins it admits the closed form

$$\begin{aligned} I_i^j = & 0.05 \ln \left(\frac{0.05}{q_{45}^j - L_*^j} \right) + 0.45 \ln \left(\frac{0.45}{q_{50}^j - q_{45}^j} \right) \\ & + 0.45 \ln \left(\frac{0.45}{q_{95}^j - q_{50}^j} \right) + 0.05 \ln \left(\frac{0.05}{U_*^j - q_{95}^j} \right) - \ln(U_*^j - L_*^j). \end{aligned} \quad (9)$$

The single information score of an expert is the average over all q questions,

$$I_i = \frac{1}{q} \sum_{j=1}^q I_i^j. \quad (10)$$

2.4 Aggregation of experts: decision makers

The two scores let us weight the experts and combine their distributions linearly into a virtual expert that we call the *decision maker* (DM). For expert i we define the combined score at level α as the product

$$CS_\alpha^i = I_i \cdot C_i \cdot \mathbf{1}(C_i \geq \alpha), \quad (11)$$

and we normalize these combined scores to obtain the weights,

$$w_i = CS_\alpha^i / \sum_{k=1}^N CS_\alpha^k. \quad (12)$$

The density of the decision maker for a question of interest j is the weighted average of the experts' densities,

$$f_{\text{PWDM}(\alpha)}^j(x) = \sum_{i=1}^N w_i f_i^j(x), \quad (13)$$

where PWDM stands for *Performance Weights Decision Maker*. A variant replaces the global information score by the information score of the specific question of interest, which gives the *Item Weights Decision Maker* (IWDM): for question j ,

$$CS_\alpha^{i,j} = I_i^j \cdot C_i \cdot \mathbf{1}(C_i \geq \alpha), \quad w_i^j = CS_\alpha^{i,j} / \sum_{k=1}^N CS_\alpha^{k,j}, \quad f_{\text{IWDM}(\alpha)}^j(x) = \sum_{i=1}^N w_i^j f_i^j(x). \quad (14)$$

A third option is the *Equal Weights Decision Maker* (EWDM), which gives every expert the same weight $1/N$ regardless of performance.

Both performance-based decision makers depend on the threshold α , so a natural choice is to optimize it. Treating the decision maker as if it was an additional expert and writing $CS(\text{DM}(\alpha))$ for its own combined score, the optimized decision maker uses

$$\hat{\alpha} = \arg \max_{\alpha \geq 0} CS(\text{DM}(\alpha)), \quad (15)$$

namely the threshold for which the decision maker performs best on the calibration questions in terms of the combined calibration-information score.

2.5 The Classical Model in practice

The reference software for the Classical Model is Excalibur, developed at TU Delft. Excalibur provides a graphical interface and computes the calibration and information scores, the performance-based weights and the optimized decision maker. It makes a study easier to reproduce and to compare with the literature. Its main practical drawback is that it only runs on Windows. Since we work on macOS, we could not use Excalibur, and we carried out the analysis instead with the R implementation of the Classical Model made available for the course (originally coded by T. Nane and T. Dong), run locally. This implementation follows exactly the formulas presented in this chapter, so the two routes are equivalent in their output; our own contribution was limited to the data-reading step, where we adapted the script to our survey exports by converting decimal commas to points and checking that the three quantiles of every assessment are strictly increasing before scoring.

It is worth explaining what the code does, since this is what gives the analysis its empirical control. For each expert it reads the 5%/50%/95% quantiles of every question and classifies each realization into one of the four inter-quantile bins to form the empirical vector s ; the calibration score is then $1 - \chi_3^2(2mI(s, p))$, exactly as in Section 2.2. The information score is the closed-form expression of Section 2.3, evaluated on a single-question intrinsic range built from the lowest 5% and the highest 95% quantile across experts (and the realization, for the calibration questions) with an overshoot $k = 0.1$. The decision maker is constructed by weighted linear aggregating. This is, the code interpolates each expert’s piecewise-uniform cumulative distribution, averages them with the chosen weights on a dense grid, and reads off the aggregate 5%/50%/95% quantiles.

2.6 Out-of-sample validation

Because the calibration questions are used both to determine expert weights and to evaluate the performance of the DM, the resulting in-sample combined score is optimistic. A common approach to moderating this bias is leave-one-seed-out cross-validation [3]. Under this procedure, one calibration question is omitted, expert weights are recomputed using the remaining questions, and the DM is then evaluated on the omitted question. This process is repeated for each calibration question in turn. Although the limited number of calibration questions in our study restricts the statistical resolution of this validation procedure, we return to it in Section 6 as a limitation and a recommendation for future work.

3 Elicitation

This chapter presents the elicitation. We first introduce the panel of experts and the data used to build the calibration questions, then we describe the elicitation protocol, and finally we list and discuss the calibration questions and the questions of interest. A characteristic of this study is that it involves two domains, namely quantum technology and financial markets, and this is reflected both in the composition of the panel and in the choice of questions.

3.1 Experts

Because the questionnaire includes two different domains, we asked two background groups of four assessors each: a *quantum group*, whose members work/study close to quantum science, and an *investment group*, whose members work close to finance and markets. All participants are quantitatively trained, which matters in an elicitation that asks them to reason in terms of quantiles and confidence intervals. Following the convention of the Classical Model, the experts are introduced here by name, but they are referred to anonymously (Expert 1 to Expert 8) in the analysis of Section 4; the quantum group corresponds to Experts 2, 4, 5 and 6, and the investment group to Experts 1, 3, 7 and 8. Individual assessments are kept confidential and are used only for academic purposes, as stated in the elicitation protocol.

To test how sensitive the conclusions are to the background of the assessors, we run the Classical Model on three panels: the **full panel** (all eight experts), the **quantum panel** (the four quantum experts) and the **investment panel** (the four investment experts). The same fourteen calibration questions and five questions of interest have been used, so the panels are directly comparable; the three sets of results are reported and contrasted in Section 4.

The panel consists of eight assessors. The academic backgrounds are concentrated in Engineering Physics, Mathematics and Engineering, which is well suited to the technical nature of the questions but also makes the group homogeneous.

3.1.1 Quantum panel

- **Bruno Casenave** is a double-degree student in Mathematics and Engineering Physics, and is about to start his bachelor's thesis at Columbia University doing research on quantum computing. He combines a strong theoretical background with direct research exposure to the field.
- **Alex Navarro** is an Engineering Physics student who will start a master's in Quantum Science and Technology next year.
- **Aniol Anglada** is also an Engineering Physics student who will start a master's in Quantum Science and Technology next year.
- The fourth member of the quantum panel was introduced to the study by the course instructor, G. F. Nane. He has expertise in the quantum domain and was included for his subject-matter experience rather than for a student profile.

3.1.2 Investment panel

- **Eva Rodriguez** is an Engineering Physics student who will start a master's in quantum computing next year. However, she was placed in the investment panel rather than the quantum one, because she works as a junior quant at Deloitte and is interested in investment. Her profile shows that the two domains of this study are not as far apart as they may seem.

- **Marc Rodriguez** is an Engineering Physics student interested in investing, who will start a master's in Mathematical Finance next year.
- **Oriol Raventos** is a double-degree student in Mathematics and Business Administration, with an interest in mathematical finance and investment.
- **Marc Gonzalez** is an Industrial Engineering² student who works part-time at KPMG in the private-equity division.

3.2 Elicitation protocol

Assessments were collected through an online questionnaire. For every quantity each assessor supplied three numbers, the 5%, 50% and 95% quantiles, with the standard interpretation that the 90% interval $[q_{0.05}, q_{0.95}]$ should contain the truth nine times out of ten, and that the median should be equally likely to sit above or below it. All responses were checked for strict monotonicity ($q_{0.05} < q_{0.50} < q_{0.95}$) before analysis, and decimal commas were normalized to points during data cleaning.

3.3 Calibration questions

An important assumption of the Classical Model is that good performance on the calibration questions predicts good performance on the questions of interest. For this reason, we designed the calibration questions along the two domains of the study, so that each panel is tested on questions close to its expertise. Specifically, the financial market and investment calibration questions were intended to assess the investment panel, whereas the quantum calibration questions were used to evaluate the quantum panel. The seeds also recreate, on purpose, the three phases of a technology wave (research, investment and adoption), since these are exactly the dimensions along which we later ask the experts to forecast. For each calibration question we give below the exact question, the assumptions provided to the experts, the realization, and a short discussion of what the question tests and how the experts answered it.

Calibration question 1: *The NASDAQ Composite Index stood at 751 points in January 1995. During the dot-com boom, the rapid growth of the World Wide Web attracted enormous investment in internet-related companies, driving the index to high values. What was the closing value of the NASDAQ Composite Index on March 10, 2000, the day it reached its all-time peak?*

– Express your answer in index points (e.g. 3000 means 3,000 points)

Realization: 5,048.62 points [10].

This question is aimed at the investment panel and tests how well the experts know the magnitude that investor enthusiasm can reach at the very top of a real technology boom. It is the natural reference point for the market question of interest (QoI 1), because an expert who knows how large the dot-com peak was should be better placed to reason about how large a future quantum market could become. It also pairs with calibration question 2 to describe the full boom-and-crash cycle. In practice, the whole panel split for this question. Five of the eight intervals contained the true peak, but Experts 1, 4 and 7 placed their whole range below it, underestimating how high the bubble reached. Investment Expert 8 was almost exact (median 5,000 against 5,048), while the quantum experts mostly covered it only through very wide ranges (Expert 5 spanning 0 to 100,000).

Calibration question 2: *After reaching its peak in March 2000, the NASDAQ crashed dramatically as investors realized most dot-com companies had no possibility to make profits. The index bottomed*

²In the Spanish system, the Industrial Engineering degree (*Ingeniería Industrial*) is a broad, generalist engineering program that combines mechanical, electrical, electronic and energy engineering with operations research and industrial management.

out on October 9, 2002, having fallen from its peak to its lowest value. By what percentage did the NASDAQ decline from its March 2000 peak to this October 2002 lowest point?

- Express your answer as a percentage decline (e.g. 50 means the index lost half of its peak value).

Realization: a 78% decline [13].

This seed complements the previous one. Question 1 measures the size of the boom while question 2 measures how violent the correction can be. It is again an investment-panel question, and it is relevant to QoI 1 because a market that can grow very fast can also fall very fast. An expert who takes into account this should produce more honest and wider intervals for the future of the quantum market. Five of the eight experts covered the 78% drop. Investment Expert 8 was again the most accurate (median 76%), whereas Experts 1 and 2 badly underestimated the true value (medians 30% and 10%) and Expert 6 appears to have misread the units, answering in the thousands.

Calibration question 3: *The World Wide Web was invented in 1991. In 1995, there were approximately 16 million internet users worldwide, which is less than 1% of the global population. Despite the dot-com crash, internet adoption continued to grow. How many internet users (in millions) were there worldwide at the end of the year 2000, when the dot-com bubble peaked?*

- Express your answer in millions of users.

Realization: approximately 361–407 million [7]. We have selected 380 million as our ‘true’ value.

This question tests the experts’ notion of the scale of adoption of an earlier technology wave, and of how quickly that scale can grow once a technology takes off. Together with calibration question 4 it forms the “adoption” block of the calibration questions, which is the block most relevant to the Fortune 500 adoption question (QoI 5). Adoption was widely underestimated: four experts (1, 2, 3 and 4) placed their whole interval below 380 million. Only Experts 5–8 covered it, with investment Expert 8 being the closest (median 320 million).

Calibration question 4: *ChatGPT was launched publicly in November 2022. Its adoption was unprecedented in speed. For example, TikTok took about 9 months to reach 100 million users after its international launch, and Instagram took about 2.5 years. How many months did ChatGPT take to reach 100 million monthly active users?*

- Express your answer as a number of months.

Realization: about 2 months [6].

This is the most recent and most extreme adoption data point available, and it tests whether the experts are aware of how fast a modern technology can grow. It informs QoI 5 directly and also has some relation with QoI 1, since revenue tends to follow adoption. This was the best understood adoption seed. In total, six of the eight covered it, and Experts 5 and 8 hit the median exactly (2 months). Only Experts 1 and 6 overestimated, treating ChatGPT’s growth as fast instead of instantaneous.

Calibration question 5: *An AI-related publication is defined as any peer-reviewed paper, conference proceeding, or preprint whose primary topic is artificial intelligence, machine learning, or related subfields, as classified by the Stanford AI Index methodology. AI-related scholarly publications have grown rapidly. In 2013, there were approximately 102,000 AI publications globally across all venues (journals, conferences, preprints). Since then, publication activity has grown every year. How many AI-related publications were produced worldwide in 2023?*

- Express your answer in thousands (e.g. 200 means 200,000 papers).

Realization: Around 242 thousand [12].

This calibration question captures the research dimension of the most recent boom, and it acts as a reference for the quantum research question (QoI 2). It is primarily a quantum-panel question in the sense that the panel reasoning about future **quant-ph** output should benefit from knowing how explosively AI research has grown. This was a strong result because seven of the eight intervals

contained the value of 242 thousand. Expert 6 was both narrow and accurate (200–300, median 250), and only Expert 3 was entirely above the realization.

Calibration question 6: *AI patent activity has grown strongly since 2010. In that year, approximately 3,800 AI-related patents were filed globally. Growth has been driven mainly by China (which now holds approximately 70% of global AI patents) and the United States. How many AI patents were filed globally in 2023?*

- Express your answer in thousands.

Realization: about 122 thousand [12].

It serves as a useful reference for the quantum-patent seed (Question 10), and it helps separate experts who understand exponential technology growth from those who extrapolate linearly. This was the worst-covered seed of all. Only Expert 8’s interval (25–180) contained the true ~122 thousand value. The next-highest value was Expert 2’s 120 thousand, and most experts remained at 100 thousand or below, so the panel collectively underestimated the explosion in AI patenting.

Calibration question 7: *Private investment in generative AI has surged in recent years. In 2022, global private investment in generative AI startups was approximately USD 3.4 billion. Since then, the sector has attracted dramatically more capital each year. How much did total global private investment in generative AI reach in 2024?*

- Express your answer in billions of USD.

Realization: USD 33.9 billion [12].

This is the clearest investment-magnitude calibration question and it is inside of the expertise of the investment panel. It is the closest analogue to the market question of interest (QoI 1), because it asks the experts to quantify the money flowing into a booming technology. Although designed for the investment panel, this question caught Experts 1, 2 and 3 too low. However, interestingly, quantum Expert 4 was the closest (median 30 against 33.9) and Expert 8 was again well placed (median 27). Five of the eight covered the value.

Calibration question 8: *Quantum computing is a young field. For reference, there are over 9,000 active non-quantum supercomputers worldwide. As of early 2025, how many operational quantum computers existed worldwide? Count only functional machines (gate-based, annealing, or cloud-accessible), not classical simulators or devices still in testing phases.*

- Express your answer as a whole number of machines.

Realization: approximately 100 machines [9].

This is a core quantum-panel question, directly underlying the third question of interest (QoI 3). Note that its realization depends on whether annealers, NMR devices and cloud prototypes are included. On this core quantum seed the panel disagreed enormously, from Expert 6’s 5-15 to Expert 1’s 1,000-20,000. Four experts covered it, and investment Expert 3 matched the median exactly (100).

Calibration question 9: *The quant-ph (quantum physics) section of arXiv, which covers quantum computing and quantum information research, had approximately 4,500 paper submissions in 2016. Since then, the field has grown significantly driven by hardware milestones and rising investment. How many papers were submitted to quant-ph in 2023?*

- Express your answer in thousands (e.g. 9 means approximately 9,000 papers).

Realization: about 9 thousand [1].

This calibration question measures the research activity of the quantum field itself and is the single most direct anchor for QoI 2, which asks for the same quantity in 2035. The performance of the quantum panel on this question is, for that reason, the most informative calibration indicator for interpreting their predictions regarding QoI 2. This is the most revealing seed of the whole study.

It is the most quantum specific question and the direct anchor for QoI 2, yet *all four* quantum experts excluded the truth value. Each of the quantum experts set a lower bound of 10 thousand or higher, slightly overestimating their own field’s output. The only assessors who covered the true ~ 9 thousand were the two investment experts who reasoned more cautiously (Experts 1 and 8), with Expert 8 the closest (median 12). This is possible evidence of domain-expert overconfidence.

Calibration question 10: *Quantum technology patents worldwide grew fivefold between 2014 and 2024. In 2014, approximately 1,000 quantum technology patents were filed globally. How many quantum technology patents were filed globally in 2024?*

- Express your answer in thousands (e.g. 4 means approximately 4,000 patents).

Realization: about 5 thousand [9].

This question is a proxy for the maturity of quantum intellectual property and complements the hardware seed (question 8). It is linked to the reasoning behind both QoI 3 and the harder “Q-day” question QoI 4, since patent activity is one of the few observable signals of how fast the hardware is advancing. Two experts (2 and 6) appear to have ignored the “in thousands” instruction, returning values three orders of magnitude too high. Among the others, Experts 5 and 8 were the closest (medians 8 and 7 against the true ~ 5).

Calibration question 11: *Venture capital (VC) investment in quantum computing has been growing but remains small compared to AI. For scale: all quantum computing raised approximately USD 5 billion cumulative over all years up to end-2023. What was the total publicly announced VC investment in quantum computing companies in 2024 alone?*

- Express your answer in millions USD (e.g. 800 means USD 800 million).

Realization: about USD 1,600 million [9].

This calibration question is at the intersection of the two panels. It is an investment figure, but about the quantum field specifically. It relates to QoI 1 and tests if the experts have a calibrated sense of how small quantum investment is compared with AI. Expert 8’s median (1,600) matched the realization exactly. Experts 3 and 6 were low by several orders of magnitude, most likely confusing millions with billions, while Expert 2 was just below the truth. In total, five experts covered the value overall.

Calibration question 12: *In the first nine months of 2025 alone, quantum computing companies raised a record amount in private investment. How much did quantum computing companies raise from private investment in just the first nine months of 2025?*

- Express your answer in millions of USD.

Realization: about USD 3,770 million [12].

This calibration question captures the fast acceleration of quantum funding during 2025. Together with question 11 it tests whether the experts are following the current trend, which is exactly the momentum they have to project forward in QoI 1. A large number of the experts underestimate the 2025 funding surge. Five of the eight intervals ended entirely below the true $\sim$ USD 3.77 billion. Only Experts 1, 4 and 8 covered it.

Calibration question 13: *McKinsey & Company has projected the number of operational quantum computers that will exist worldwide by 2030. McKinsey expects rapid scaling over the next five years as hardware matures. What is the lower bound of McKinsey’s projected range for the number of operational quantum computers by 2030?*

- Express your answer as a number of machines.

Realization: 2,000 machines [8].

This question tests if the experts are aware of published expert projections for the field, and it relates directly to QoI 3. It is interesting because it asks the experts to recall someone else’s forecast rather

than a historical fact. Five of the eight covered McKinsey’s lower bound, and quantum Expert 5 hit the median exactly (2,000). Experts 1, 2 and 7 set their 95% quantiles below the true figure, again underestimating the projected pace of hardware scaling.

Calibration question 14: *According to McKinsey’s annual AI survey (cited in the Stanford AI Index 2025), in 2023 approximately 55% of organisations reported using AI in at least one business function. What percentage of organisations reported doing so in 2024?*

- Express your answer as a percentage from 0 to 100.

Realization: 78% [12].

This is the business-adoption calibration question and the natural analogue of QoI 5, which asks the same kind of question for quantum computing in 2035. Supported by the adoption-speed seeds 3 and 4, it tests whether the experts can quantify how quickly a technology spreads through real companies. This seed was well understood. In total, six of the eight covered it, with most medians clustered at 75–80% against the true 78%. Expert 3 was too high and Expert 6 was the clear outlier (10–30%).

3.4 Questions of interest

The questions of interest are the actual targets of the study. They are forecasts about the future of quantum computing for which no reliable data exist. Each of them was linked, during the design of the elicitation protocol, with one or more calibration questions. We make these relations explicit below.

Question of interest 1: *The total quantum computing market was estimated at USD 1–3 billion in 2024. For comparison, the AI software market reached USD 50 billion within roughly 10 years of the deep-learning breakthrough in 2012. In what year will the global quantum computing market first exceed USD 50 billion in annual revenue?*

- Express your answer as a year; if you believe it will not happen before 2060, use 2060 as your 95th percentile.

This question concerns the speed of commercial growth, and it is the forecast for which the investment panel is most relevant. It draws on the investment block of the calibration questions: the magnitudes of generative-AI investment (seed 7) and of current quantum venture capital (seeds 11 and 12) set the starting scale, while the dot-com boom and crash (seeds 1 and 2) show the experts how quickly such markets can both rise and fall.

Question of interest 2: *How many papers per year, in thousands, will be submitted to *quant-ph* by 2035?*

- Express your answer in thousands (e.g. 30 means 30,000 papers per year).

This is a direct extrapolation of seed 9, which asks for the same quantity in 2023, with seed 5 (AI publications) acting as a fast growing reference for what an explosive research field looks like. The performance of the quantum panel on seed 9 is therefore the most informative calibration signal for this forecast, and we expect the quantum panel to dominate the decision maker here.

Question of interest 3: *How many operational quantum computers will exist worldwide in 2035?*

- Express your answer as a number of machines.

This forecast builds directly on seed 8, the 2025 baseline, and on seed 13, McKinsey’s lower-bound projection for 2030. Together they bracket the trajectory that the experts must extend to 2035.

Question of interest 4: *RSA-2048 encryption secures most internet traffic today. Breaking it is estimated to require millions of error-corrected logical qubits, which is far beyond current capabilities, where advanced processors reach thousands of physical qubits. By what year will a quantum computer first break RSA-2048 encryption within 24 hours under realistic operational conditions?*

- Express your answer as a year; if you believe it will not happen before 2070, use 2070 as your 95th percentile.

This is the well-known “Q-day” question and the one that captures the gap between current hardware and a disruptive milestone. It is the hardest question to anchor because it has no calibration question analogue. The closest ones are the hardware-maturity calibration questions 8 and 10. We explain this missing link explicitly as a limitation in Section 6, since it means the forecast for QoI 4 has a weaker calibration than the others.

Question of interest 5: *Quantum adoption is expected to lag behind AI adoption significantly, as the hardware is far less mature. What percentage of Fortune 500 companies will report operational use of quantum computing by 2035?*

- Express your answer as a percentage from 0 to 100.

This is a business-adoption forecast whose natural calibration analogue is seed 14 (organizations using AI in 2024), supported by the adoption-speed seeds 3 and 4. Because adoption follows the maturity of the underlying hardware, this question is also implicitly tied to the hardware forecasts QoI 3 and QoI 4.

3.5 Data information

The true values of the calibration questions were taken from public and citable sources, which we list here so that the calibration questions can be reproduced and checked. The two dot-com questions (questions 1 and 2) rely on the historical levels of the NASDAQ Composite [10] and on the account of the dot-com bubble in [13]: a closing peak of 5,048.62 points on 10 March 2000 and a fall of about 78% to the October 2002 trough³. The two adoption seeds use Internet Live Stats for the number of internet users at the end of 2000 [7] and a Reuters analyst note for the speed at which ChatGPT reached 100 million users [6].

The block of seeds that benchmarks the present wave against artificial intelligence comes mainly from the Stanford AI Index. The 2025 edition [12] provides the number of AI publications in 2023, the private investment in generative AI in 2024 and the share of organizations using AI in 2024, while the 2024 edition [11] provides the AI patent count. The quantum seeds are based on the MIT Quantum Index Report 2025 [9], which gives the number of operational quantum computers, the quantum patents and the quantum venture capital in 2024. They are also based on the arXiv statistics for the `quant-ph` submissions [1] and on McKinsey’s projection to 2030 [8]. The amount that quantum companies raised from private investment during the first nine months of 2025 was taken from the industry tracker AIMultiple.

³The lowest turning point of a business cycle, marking the end of a recession before economic recovery begins.

4 Analysis: expert and decision-maker performance

In this chapter the uncertainty assessments are analyzed with the Classical Model. Because Excalibur runs only on Windows, the analysis was carried out with the R implementation of the Classical Model provided for the course (originally coded by T. Nane and T. Dong), run locally on macOS. This implementation follows exactly the formulas of Section 2, so the two routes are equivalent in their output. We first state the analysis settings, then, we discuss the performance of the experts, and finally the performance of the decision makers, for the full panel and the two background sub-panels.

4.1 Analysis settings

Four settings have to be fixed before scoring, exactly the ones Excalibur gives: the way the weights are chosen, the optimization of the threshold α , the calibration power and the intrinsic range.

Choice of weights. The Classical Model can build the decision maker with equal weights, global (performance) weights, item weights or user weights. We use the *equal-weight* decision maker (EWDM) and the *global performance-weight* decision maker (PWDM); the latter is the choice that corresponds exactly to the Classical Model of Section 2.4. Item weights (IWDM), which recompute the information score per question, are not considered. User weights are not considered, as no self-assessment of expertise was collected and such weights are prone to overconfidence biases that the method is designed to avoid.

Optimization of α . The performance-based decision makers depend on the threshold α , the minimum calibration score an expert must reach to receive a non-zero weight. We report the non optimized case $\alpha = 0$ (every expert contributes in proportion to $C(e)I(e)$) and the optimized case $\hat{\alpha} = \arg \max_{\alpha} CS(DM(\alpha))$, in which α is chosen so that the decision maker’s own combined score is maximal.

Calibration power. Excalibur multiplies the statistic by a calibration power in $[0.1, 1]$; setting it to 1 recovers the calibration score of Equation (4). We use power = 1 throughout, so our calibration scores are directly comparable with the theory.

Intrinsic range. The information score requires a bounded support. We use the conventional $k = 10\% = 0.1$ overshoot on the smallest interval containing all experts’ quantiles and the realization, with a uniform background measure. For the questions expressed as percentages the natural support is $[0, 100]$. We note, however, the limitation that the same overshoot rule is nonetheless applied to questions spanning several orders of magnitude. This is discussed in Section 6.

4.2 Performance of the experts

Tables 2–4 report, for the three panels, the calibration score $C(e)$, the mean relative information over all nineteen questions (“total”) and over the fourteen calibration questions only (“realization”), the non-normalized weight $C(e) \cdot I_{\text{real}}(e)$, and the normalized weights without and with the decision maker included. The layout follows the format of Excalibur output. The last row of each table is the resulting $\alpha = 0$ performance-weight decision maker.

Table 2: Full panel (8 experts): expert performance for the global-weight decision maker with $\alpha = 0$.

Id	Calibration score	Mean rel. total	Mean rel. realization	Unnorm. weight	Norm. weight without DM	Norm. weight with DM
Exp1	1.01×10^{-4}	1.699	1.904	1.93×10^{-4}	1.29×10^{-4}	9.45×10^{-5}
Exp2	4.87×10^{-6}	2.018	2.213	1.08×10^{-5}	7.20×10^{-6}	5.27×10^{-6}
Exp3	1.29×10^{-6}	2.689	3.031	3.90×10^{-6}	2.61×10^{-6}	1.91×10^{-6}
Exp4	1.57×10^{-4}	1.790	2.056	3.22×10^{-4}	2.15×10^{-4}	1.58×10^{-4}
Exp5	0.4312	1.514	1.629	0.7023	0.4695	0.3436
Exp6	1.06×10^{-6}	2.281	2.548	2.70×10^{-6}	1.81×10^{-6}	1.32×10^{-6}
Exp7	7.20×10^{-4}	1.921	2.092	1.51×10^{-3}	1.01×10^{-3}	7.37×10^{-4}
Exp8	0.3994	1.785	1.981	0.7913	0.5291	0.3872
PWDM($\alpha=0$)	0.3994	1.212	1.373	0.5483	—	0.2683

Table 3: Quantum panel (Experts 2, 4, 5, 6): expert performance, global weights, $\alpha = 0$.

Id	Calibration score	Mean rel. total	Mean rel. realization	Unnorm. weight	Norm. weight without DM	Norm. weight with DM
Exp2	4.87×10^{-6}	1.666	1.815	8.83×10^{-6}	1.66×10^{-5}	8.39×10^{-6}
Exp4	1.57×10^{-4}	1.430	1.648	2.58×10^{-4}	4.86×10^{-4}	2.45×10^{-4}
Exp5	0.4312	1.168	1.230	0.5303	0.9995	0.5041
Exp6	1.06×10^{-6}	1.921	2.142	2.27×10^{-6}	4.28×10^{-6}	2.16×10^{-6}
PWDM($\alpha=0$)	0.4312	1.136	1.209	0.5214	—	0.4957

Table 4: Investment panel (Experts 1, 3, 7, 8): expert performance, global weights, $\alpha = 0$.

Id	Calibration score	Mean rel. total	Mean rel. realization	Unnorm. weight	Norm. weight without DM	Norm. weight with DM
Exp1	1.01×10^{-4}	0.818	0.986	1.00×10^{-4}	2.38×10^{-4}	1.29×10^{-4}
Exp3	1.29×10^{-6}	1.784	2.095	2.70×10^{-6}	6.41×10^{-6}	3.48×10^{-6}
Exp7	7.20×10^{-4}	1.041	1.182	8.51×10^{-4}	2.02×10^{-3}	1.10×10^{-3}
Exp8	0.3994	0.894	1.052	0.4203	0.9977	0.5424
PWDM($\alpha=0$)	0.3565	0.842	0.992	0.3536	—	0.4563

Three observations follow from these tables, visualized in Figure 2:

1. **Calibration is concentrated in two experts.** Across the full panel only **Expert 5** ($C = 0.431$, quantum group) and **Expert 8** ($C = 0.399$, investment group) reach a calibration score above the conventional 0.05 threshold; the other six lie between 10^{-6} and 10^{-3} , which means that their realizations are extremely unlikely under their stated distributions. Because the weight is the product $C(e)I(e)$, these two experts get almost all of the performance weight (0.47 and 0.53 without the DM).
2. **Information does not track calibration.** Notice that the most informative experts are not the best calibrated. Expert 3, for instance, has the highest mean relative information (3.03 on the realizations), however, has the worst calibration score (1.3×10^{-6}). Narrow intervals that repeatedly miss the truth are penalized exactly as the model intends.

3. **One expert dominates each sub-panel.** In the quantum panel Expert 5 carries 99.9% of the weight, and in the investment panel Expert 8 carries 99.8%. The four-expert sub-panels therefore effectively reduce to a single assessor, which matters for the interpretation of their forecasts.

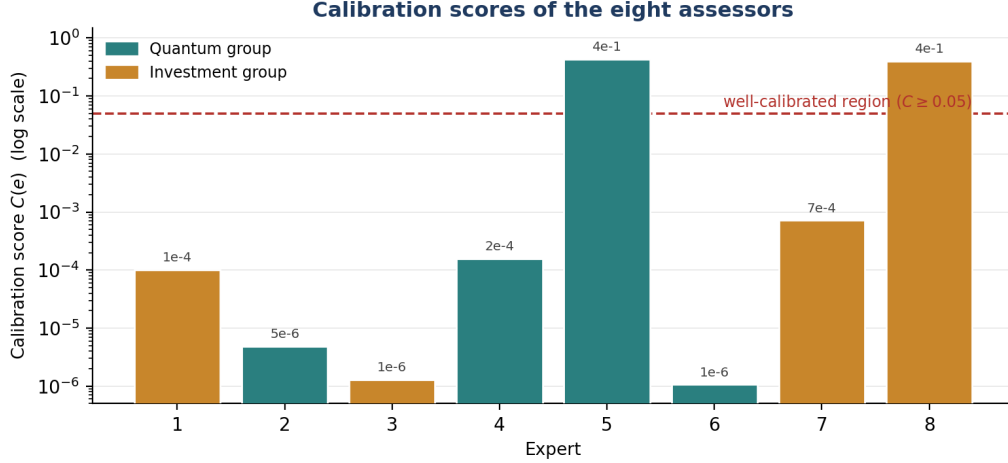


Figure 2: Calibration scores of the eight assessors on a logarithmic scale. Only Experts 5 (quantum) and 8 (investment) exceed the conventional well-calibrated threshold ($C \geq 0.05$, dashed line); the other six are overconfident by one to six orders of magnitude. Color denotes background group.

4.2.1 Empirical coverage and domain-expert overconfidence

The calibration score summarizes an expert’s performance through a single p-value. As a complementary, distribution-free diagnostic, we also examine the empirical coverage, defined as the proportion of seed realizations that actually lie within the expert’s stated 90% interval $[q_{0.05}, q_{0.95}]$. A perfectly calibrated assessor covers about 90% of the calibration questions. Table 5 reports the coverage of each panel, split into the six quantum-specific seeds (CAL08-CAL13) and the eight financial / adoption seeds.

Table 5: Empirical coverage: fraction of seed realizations inside the experts’ 90% intervals, by group and seed domain. A well-calibrated assessor would cover $\approx 90\%$.

Group	Quantum seeds (6 per expert)	Financial / adoption (8 per expert)	All seeds (14 per expert)
Quantum (Exp 2,4,5,6)	7/24 = 29.2%	20/32 = 62.5%	27/56 = 48.2%
Investment (Exp 1,3,7,8)	14/24 = 58.3%	19/32 = 59.4%	33/56 = 58.9%

The result gives some valuable insights. On the financial and adoption seeds the two groups are comparable (62.5% vs 59.4%). On the six *quantum* seeds, however, the investment group covers more than twice as many realizations as the quantum group (58.3% vs 29.2%). Note that the domain experts were the least accurate precisely on their own subject. The clearest case is calibration question 9 (**quant-ph** submissions), where all the four quantum experts placed their lower bound at or above 10,000 and thus excluded the true value of $\sim 9,000$, while the generalist Expert 8 covered it comfortably. This is empirical evidence of domain-expert overconfidence, and it explains a result that

would otherwise look unimaginable in the next subsection, which is that the quantum panel’s *equal-weight* decision maker is better calibrated than its performance-weight one, because the physicists’ individually overconfident intervals only cover the truth once their disagreement is aggregated.

4.3 Performance of the decision makers

Table 6 compares the equal-weight and the $\alpha = 0$ performance-weight decision makers for the three panels, on calibration, information (over all questions and over the calibration questions), and the combined score $C \cdot I_{\text{real}}$ that is used to rank decision makers.

Table 6: Decision-maker performance for the three panels. The combined score $C \cdot I_{\text{real}}$ is the ranking criterion.

Panel	DM	Calibration	Info. (all)	Info. (cal.)	Combined
Full (8 experts)	PWDM($\alpha=0$)	0.399	1.212	1.373	0.548
	EWDM	0.132	0.740	0.822	0.109
Quantum (4 experts)	PWDM($\alpha=0$)	0.431	1.136	1.209	0.521
	EWDM	0.569	0.504	0.529	0.301
Investment (4 exp.)	PWDM($\alpha=0$)	0.357	0.842	0.992	0.354
	EWDM	0.357	0.310	0.382	0.136

Note that performance weighting beats equal weighting in every panel. On the combined score the PWDM dominates the EWDM for the full panel (0.548 vs 0.109), the quantum panel (0.521 vs 0.301) and the investment panel (0.354 vs 0.136). The gap is largest for the full panel, where equal weighting is reduced by averaging in the six overconfident experts. This reproduces the standard finding of the Classical Model literature that performance-based weights out-perform equal weights.

Also, notice that optimizing α collapses each panel onto its calibrated experts. For the full panel the optimized PWDM retains only Experts 5 and 8, for the quantum panel only Expert 5, and for the investment panel only Expert 8. For the full panel the question of interest quantiles of the optimized PWDM match those of the $\alpha = 0$ PWDM to within rounding (the full $[19 \times 3]$ EWDM and PWDM matrices are given in Appendix A). The fact that the forecasts are similar across the weighting schemes considered suggests that the results are mainly determined by the calibrated experts, rather than by the specific value chosen for α .

The three candidate decision makers are the performance-weight DMs of the full, quantum and investment panels. The investment-panel DM is the weakest on every metric (combined 0.354) and is discarded. The choice is therefore between the full panel (combined 0.548, calibration 0.399) and the quantum panel (combined 0.521, calibration 0.431). The quantum panel is marginally better calibrated, but this is achieved by a decision maker that is, in practice, a single expert (Expert 5 carries 99.9% of its weight). This panel would not be robust or diverse, and it inherits the domain-expert overconfidence documented in Section 4.2.1, where the quantum group covered only 29% of the quantum seeds. The full-panel PWDM has the highest combined score of any decision maker considered, good calibration, and the desirable property that it aggregates the two well-calibrated assessors from *different* backgrounds (Expert 5 from quantum, Expert 8 from investment), so that no single person or perspective determines the forecast. For these reasons the **full-panel performance-weight**

decision maker is selected to answer the questions of interest in Section 5. The two sub-panel decision makers are used only to expose where background changes the forecast.

5 Results: forecasts for the questions of interest

In this chapter the selected decision maker, the full-panel performance-weight DM, is used to answer the five questions of interest. For each question we give its 5%, 50% and 95% quantiles, the resulting best estimate and 90% interval, and a short interpretation. The forecasts of the two background sub-panels are reported alongside, in Table 7 and Figures 3–4, to show where the quantum and investment perspectives disagree.

Table 7: Questions of interest: performance-weight decision-maker quantiles (5% / 50% / 95%) for the three panels. The full-panel column is the recommended forecast.

Question of interest	Full panel	Quantum	Investment
QoI1 – Year market >USD 50 bn	2032 / 2039 / 2050	2033 / 2036 / 2045	2032 / 2042 / 2050
QoI2 – quant-ph 2035 (thous.)	11 / 49 / 459	30 / 100 / 500	10 / 30 / 60
QoI3 – Operational QC in 2035	609 / 3,220 / 9,968	600 / 1,001 / 2,976	2,862 / 4,996 / 9,999
QoI4 – Year RSA-2048 broken (Q-day)	2035 / 2049 / 2067	2035 / 2040 / 2060	2040 / 2055 / 2068
QoI5 – Fortune 500 quantum use 2035 (%)	5 / 14 / 49	5 / 15 / 50	7 / 12 / 28

***Question of interest 1:** In what year will the global quantum computing market first exceed USD 50 billion in annual revenue?*

The decision maker returns $(q_5, q_{50}, q_{95}) = (2032, 2039, 2050)$. The best estimate is therefore **2039**, and it is believed with 90% probability that the threshold remains in the interval [2032, 2050]. For comparison, the AI software market reached a similar scale within about a decade of the 2012 deep-learning breakthrough. The panel expects quantum to take noticeably longer because the hardware is far less mature. The quantum group is more optimistic (median 2036) than the investment group (median 2042), the first sign of a disagreement that continues through all five forecasts.

***Question of interest 2:** How many papers per year (in thousands) will be submitted to *quant-ph* by 2035?*

The decision maker returns (11, 49, 459). The best estimate is $\approx 49,000$ submissions per year, with a 90% interval of [11,000, 459,000]. The median is an order of magnitude above the $\sim 4,500$ of 2016, but the wide upper tail reflects disagreement about whether the field enters AI-style exponential growth. Here the divergence is at its sharpest point: the quantum group (median 100,000) expects far more output than the investment group (median 30,000).

***Question of interest 3:** How many operational quantum computers will exist worldwide in 2035?*

The decision maker returns (609, 3,220, 9,968). The best estimate is $\approx 3,200$ machines, with a 90% interval of [609, 9,968], roughly a twenty-fold increase over the estimated 100–200 today and still below the $>9,000$ classical supercomputers in operation. Interestingly, the divergence *changes* here: the investment group expects more machines (median $\sim 5,000$) than the quantum group (median $\sim 1,000$).

***Question of interest 4:** By what year will a quantum computer first break RSA-2048 within 24 hours under realistic conditions?*

The decision maker returns (2035, 2049, 2067). The best estimate of this “Q-day” is **2049**, with a 90% interval of [2035, 2067]. This suggests that “Q-day” is unlikely to occur within the next decade, but could plausibly take place within the operational lifetime of infrastructure deployed today. The quantum group again expects this milestone earlier (median 2040) than the investment group (median 2055). This is the least well-anchored forecast, as it has no direct calibration analogue (Section 6).

Question of interest 5: *What percentage of Fortune 500 companies will report operational use of quantum computing by 2035?*

The decision maker returns (5, 14, 49). The best estimate is $\approx 14\%$, with a 90% interval of [5%, 49%], which is below the $\sim 55\text{--}78\%$ reported for AI today, consistent with the fact that the widespread adoption of quantum technologies will occur significantly later than that of AI. The two groups broadly agree on the median ($\sim 12\text{--}15\%$), however, the investment group is the more conservative on the upper tail.

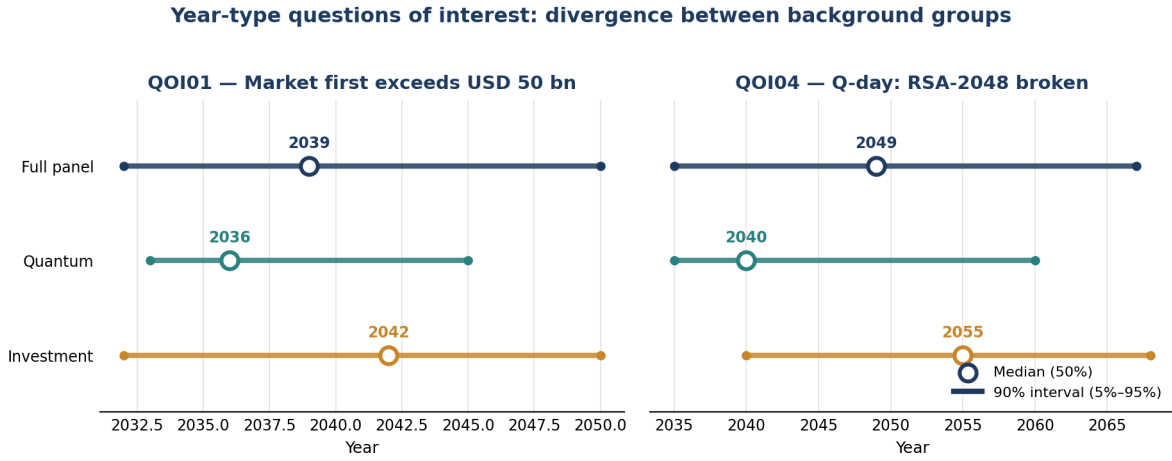


Figure 3: Year-type questions of interest. Performance-weight DM medians (open circles) and 90% intervals (bars) for the full panel and the two background groups.

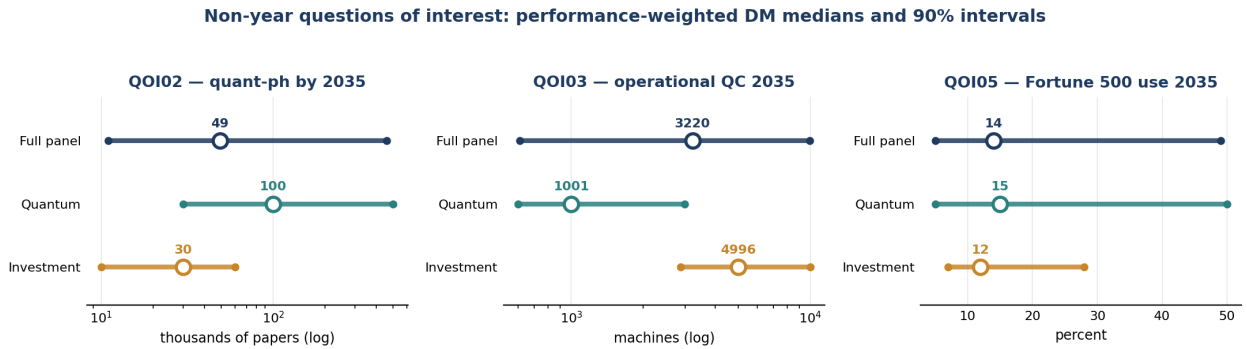


Figure 4: The three non-year questions of interest.

5.1 Benchmarking the forecasts against the historical waves

It is worth comparing these medians next to the reference quantities of Table 1. The research median of $\sim 49,000$ quant-ph papers by 2035 would put the field roughly where all of AI publishing stood

in the mid-2010s, and at about a quarter of total **arXiv** output in 2023, which is aggressive, but not implausible if quantum follows the AI trajectory. The hardware median of $\sim 3,200$ machines by 2035 is about a third of the number of today's classical supercomputers, a reasonable early-adoption quantity. The projected market milestone of USD 50 bn by 2039 suggests strong growth in the quantum computing sector, although at a slower pace than the AI software market's rise to a similar scale. This is consistent with the fact that quantum technologies rely on a less mature hardware ecosystem. Similarly, the forecast that approximately 14% of Fortune 500 companies will have adopted quantum computing by 2035 indicates a much lower level of penetration than is currently observed for AI. Taken together, these results suggest that experts expect quantum computing to remain a specialized technology over the next decade, rather than evolving into a broadly adopted general tool.

6 Discussion

This chapter makes some comments on the results and on the study itself, with the objective of showing some limitations or interesting facts to the reader.

6.1 Where the two background groups disagree

The three-panel design was introduced to test whether the background of the assessors changes the conclusions, and it does, in a certain way. The quantum specialists are consistently more aggressive on the technical quantum questions. They expect the market to cross USD 50 bn sooner (median 2036 vs 2042), far more research output (median 100k vs 30k `quant-ph` papers) and an earlier Q-day (median 2040 vs 2055). The investors are more optimistic about *deployment scale*, expecting more operational machines in 2035 (median $\sim 5,000$ vs $\sim 1,000$), but more conservative on the disruptive milestones. In short, the people closest to the physics are optimistic about capability and timing, while the people closest to the money are optimistic about magnitude but cautious about disruption. The recommended full-panel decision maker lies between the two on every question, which is the desired behavior of an aggregate forecast.

6.2 Domain expertise did not improve calibration

The most important methodological finding is that subject expertise did not translate into better calibration. The quantum specialists were, individually, among the most overconfident assessors (Experts 2 and 6 have calibration scores around 10^{-6}), and the coverage analysis of Section 4.2.1 showed that on the six quantum calibration questions the quantum group covered only 29% of the realizations against 58% for the generalist investors. The single best calibrated quantum expert (Expert 5) is what “saves” the quantum decision maker. This is consistent with a documented pattern in the Classical Model literature, which is that narrow expertise often comes with narrow, overconfident intervals, which is exactly why the method scores calibration empirically rather than trusting credentials. It also justifies, after this finding, the inclusion of the investment group.

6.3 Weak anchoring of the Q-day forecast

By construction, each question of interest was paired with one or more calibration questions of the same type, so that performance on the seeds is informative about performance on the forecast. The exception is QoI4, the Q-day question. There is no calibration question whose answer is a future hardware-capability date, so the nearest anchors are only the hardware-maturity seeds (CAL08 and CAL10). The forecast for QoI4 should therefore be read with more caution than the others, and the 90% interval ([2035, 2067]) is therefore wide.

6.4 Limitations of the study

Several limitations affect the conclusions. *(i) Panel composition.* The eight assessors were recruited from my circle and are homogeneous in training (Engineering Physics, Mathematics, Engineering). *(ii) No training.* The elicitation was conducted online without training. Had we conducted one, we could have reduced the unit errors visible in the data (for instance, two experts answered CAL10 three orders of magnitude too high, possibly having read “thousands” as units, and one answered CAL02 in the thousands). *(iii) Few seeds.* With only fourteen calibration questions the calibration p -values have limited resolution. *(iv) In-sample evaluation.* The decision maker was selected on the same calibration questions used to compute the weights. A leave-one-seed-out cross-validation

would test whether the dominance of Experts 5 and 8 is robust. *(v) Background measure.* A uniform background on a 10% overshoot was applied to every question, including the questions that span several orders of magnitude. Maybe a log-uniform measure background, as discussed in Section 2.3, would have been more appropriate for those items and would change the information scores somewhat.

6.5 A note on the χ^2 approximation

A more fundamental warning concerns the calibration score itself. The score relies on the statistic $2m I(s, p)$ being χ^2_3 -distributed, which holds only asymptotically in the number of seeds m . In this study we had $m = 14$ calibration questions, and for such a small m the approximation may be poor. [5] show that the number of seed variables has a material effect on the performance of the Classical Model, and [2] discusses the limitations of interpreting the calibration p -value as an exact measure of an expert's calibration. This does not affect the *ranking* of the experts (Experts 5 and 8 are clearly the best calibrated under any reasonable scoring), but it is a reason not to over interpret the precise values.

6.6 Recommendations for future work

A next study should recruit a more varied expert panel (e.g. PhD students or more experienced experts). Several improvements could strengthen future elicitations. First, conducting a pilot elicitation would help identify and resolve any ambiguities before the main study. Second, a larger set of seed questions, ideally balanced across the relevant domains of expertise. Third, log-uniform background measures could be used for questions involving estimates across multiple orders of magnitude. Finally, the robustness of the decision maker could be assessed through a leave-one-seed-out cross-validation analysis. With those changes the same three-panel design would give a better result of how domain background shapes forecasts of the quantum-computing boom.

7 Conclusion

This study used the Classical Model, implemented in R, to quantify and aggregate the uncertainty of eight assessors about the future of quantum computing, and it compared decision makers built from the full panel and from two background sub-panels. Performance weighting outperformed equal weighting in every configuration on the combined calibration and information score. Of the three candidate decision makers the full-panel performance weight DM was selected, because it has the highest combined score (calibration 0.399, information 1.37, combined 0.548) and, unlike the sub-panel decision makers that each collapse onto a single expert, it aggregates the two well-calibrated assessors from different backgrounds.

According to this decision maker, the global quantum-computing market will first exceed USD 50 billion around **2039** (90% interval 2032-2050), there will be roughly **3,200** operational quantum computers by 2035, RSA-2048 will first be broken within 24 hours around **2049** (90% interval 2035-2067), and only about **14%** of Fortune 500 companies will report operational use of quantum computing by 2035.

Beyond the numerical forecasts, the most notable finding is methodological. Experts' professional backgrounds appear to influence their predictions in a systematic way. Physicists tended to be more optimistic about technical progress and timelines, whereas investors were generally more optimistic about the scale of future deployment. However, neither group's confidence was reflected in its calibration performance. The coverage analysis made this visible. On their own quantum seeds the physicists covered barely a quarter of the realizations, far fewer than the generalist investors. This finding highlights the value of the Classical Model's performance-based approach to expert weighting. By evaluating experts on the basis of calibration instead of their credentials alone, the model identified a well-calibrated decision maker.

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A Full decision-maker solutions

The tables below give the full $[19 \times 3]$ decision-maker matrices (5% / 50% / 95% quantiles) returned by the R pipeline, for the equal-weight (EWDM) and performance-weight (PWDM) decision makers of each of the three panels. Rows CAL01–CAL14 are the calibration questions and QOI01–QOI05 the questions of interest.

Table 8: Full panel (8 experts) — EWDM and PWDM.

ID	EWDM			PWDM		
	5%	50%	95%	5%	50%	95%
CAL01	828.45	4604.87	74589.49	1233.25	5782.27	94650.13
CAL02	5.69	70.32	4382.53	50.02	78.33	106.72
CAL03	30.56	107.19	808.96	35.07	257.56	596.09
CAL04	1.00	3.96	15.12	1.00	2.00	6.89
CAL05	126.28	337.71	1094.17	103.64	238.69	496.18
CAL06	6.17	54.71	141.13	2.29	53.36	172.00
CAL07	4.17	24.54	326.01	15.08	36.21	458.74
CAL08	5.22	59.12	13087.52	19.84	96.69	299.97
CAL09	7.52	21.37	88.63	7.10	18.41	93.84
CAL10	3.10	13.61	5423.19	2.93	7.52	38.32
CAL11	5.12	630.45	7574.14	131.99	1318.35	3949.79
CAL12	3.77	546.86	4293.90	133.39	1487.69	4774.96
CAL13	13.12	1153.48	8753.85	812.03	3370.31	9619.46
CAL14	15.98	76.52	93.14	60.00	77.00	94.39
QOI01	2028.68	2034.79	2046.73	2032.10	2039.01	2049.50
QOI02	8.09	48.90	335.08	10.50	49.24	459.07
QOI03	377.27	2382.10	74319.41	609.08	3220.05	9967.62
QOI04	2029.88	2039.45	2062.40	2035.18	2048.98	2067.23
QOI05	6.04	23.38	84.58	5.21	13.62	48.67

Table 9: Quantum panel (Experts 2, 4, 5, 6) — EWDM and PWDM.

ID	EWDM			PWDM		
	5%	50%	95%	5%	50%	95%
CAL01	790.34	5262.76	87048.78	0.21	49973.42	99997.45
CAL02	5.57	90.43	4714.88	49.84	90.00	100.42
CAL03	28.48	92.39	472.96	29.96	149.94	499.98
CAL04	0.82	4.70	16.55	1.00	2.00	7.02
CAL05	115.63	296.18	821.47	100.00	200.15	501.12
CAL06	4.26	37.77	112.37	1.00	20.04	100.00
CAL07	4.25	26.43	404.34	19.84	99.96	499.98
CAL08	5.63	31.49	174.03	19.85	49.99	150.00
CAL09	10.12	21.25	85.91	10.00	39.98	100.00
CAL10	3.91	1815.46	5753.96	3.00	8.00	19.38
CAL11	5.45	681.67	8714.96	100.01	950.49	2068.27
CAL12	105.97	406.67	994.12	100.00	500.00	805.84
CAL13	11.41	958.30	8775.42	799.59	2001.03	5026.08
CAL14	12.71	71.25	92.58	60.00	80.00	95.00
QOI01	2030.08	2034.38	2043.27	2032.97	2036.00	2045.00
QOI02	18.15	94.61	410.97	29.84	99.97	499.98
QOI03	624.77	2226.66	86921.06	600.01	1000.53	2975.56
QOI04	2030.79	2038.27	2055.81	2034.96	2040.00	2060.00
QOI05	5.52	18.39	43.55	5.00	15.01	50.00

Table 10: Investment panel (Experts 1, 3, 7, 8) — EWDM and PWDM.

ID	EWDM			PWDM		
	5%	50%	95%	5%	50%	95%
CAL01	903.78	4155.58	17489.56	3982.32	4998.17	5999.97
CAL02	21.31	61.96	110.37	59.13	76.00	85.62
CAL03	33.13	118.53	901.57	89.50	320.46	608.80
CAL04	1.08	3.65	9.19	1.00	2.00	4.12
CAL05	156.21	371.10	1148.39	150.00	250.13	409.47
CAL06	12.23	61.29	157.27	24.89	69.97	179.98
CAL07	4.36	23.22	89.90	14.93	27.02	42.02
CAL08	10.62	112.54	16304.23	21.54	129.86	393.36
CAL09	7.10	21.50	89.88	7.00	12.02	24.70
CAL10	2.79	11.84	43.05	3.00	7.01	14.08
CAL11	5.05	487.24	3747.92	658.82	1598.06	3999.51
CAL12	3.09	1378.86	4755.04	1358.58	2299.33	4999.95
CAL13	321.45	1234.68	8728.55	964.79	4990.54	9998.87
CAL14	61.21	80.83	93.15	60.00	75.02	90.05
QOI01	2028.28	2035.52	2048.21	2031.92	2041.98	2050.00
QOI02	7.20	35.83	58.52	9.94	29.99	60.00
QOI03	420.36	2509.81	9539.96	2861.77	4995.80	9999.00
QOI04	2029.07	2042.00	2065.04	2039.68	2054.97	2068.00
QOI05	7.36	40.12	87.72	7.00	12.03	27.87

B Calibration question plots

For all experts the uncertainty quantification is visualized by means of using a line between the 5% and 95% quantile. Their 50% quantile estimation is visualized by a dot. The blue dashed line represents the true realization of the calibration question

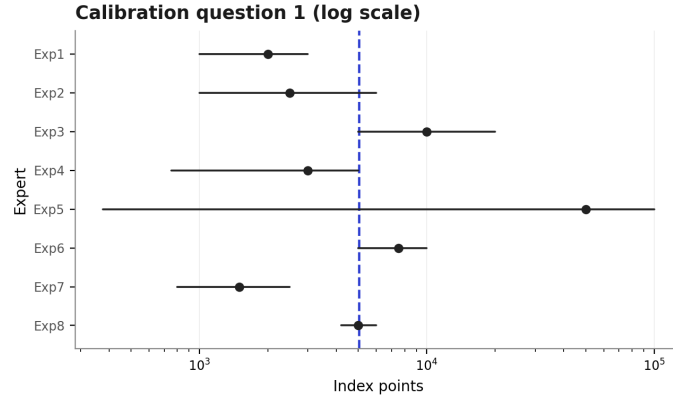


Figure 5: Range plot of calibration question 1.

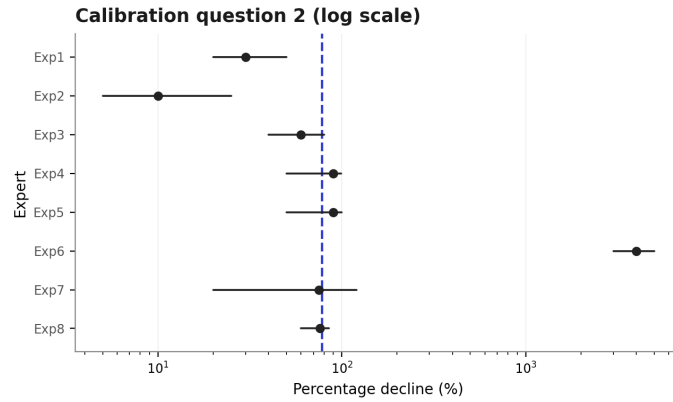


Figure 6: Range plot of calibration question 2.

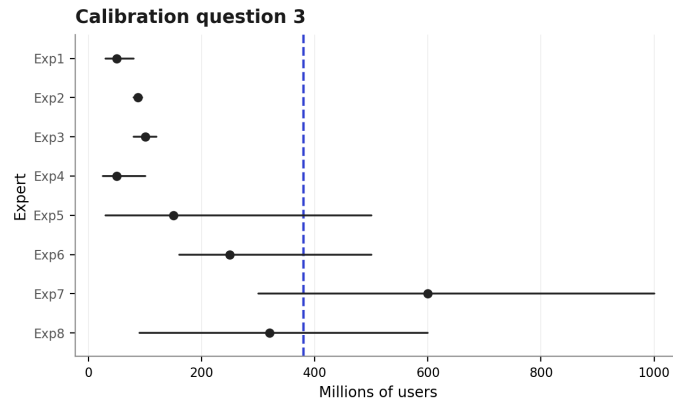


Figure 7: Range plot of calibration question 3.

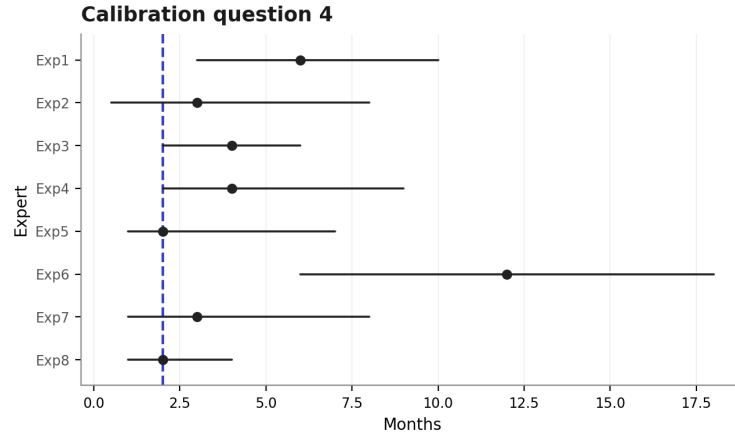


Figure 8: Range plot of calibration question 4.

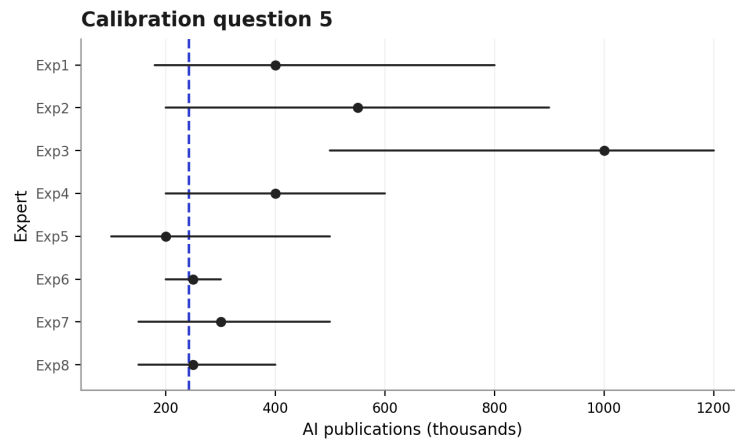


Figure 9: Range plot of calibration question 5.

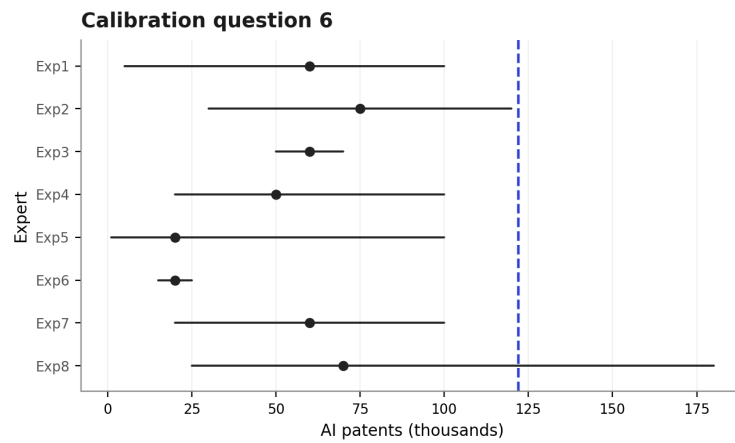


Figure 10: Range plot of calibration question 6.

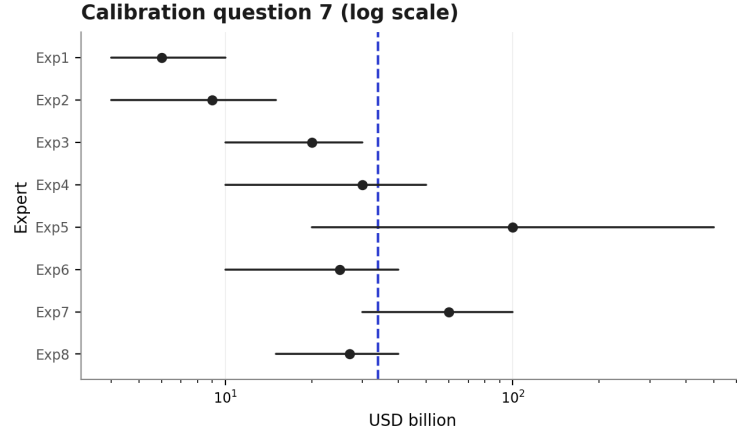


Figure 11: Range plot of calibration question 7.

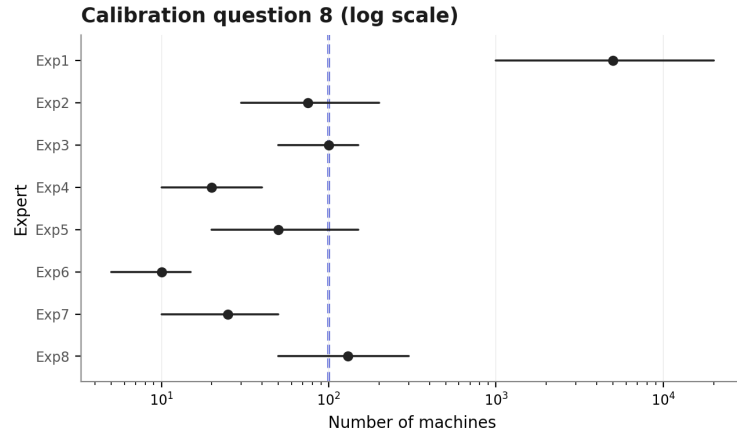


Figure 12: Range plot of calibration question 8.

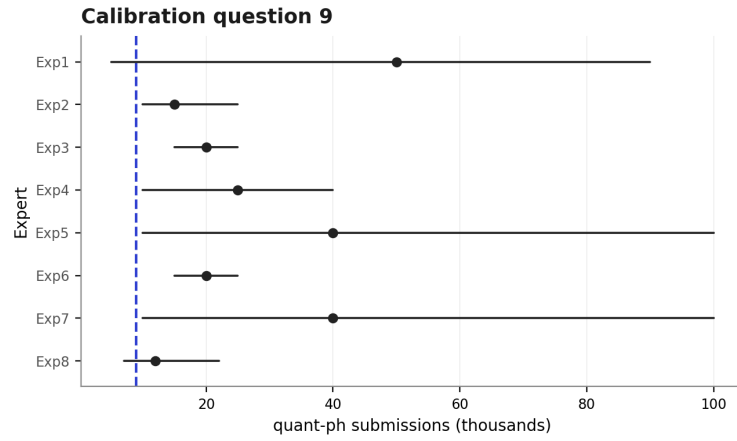


Figure 13: Range plot of calibration question 9.

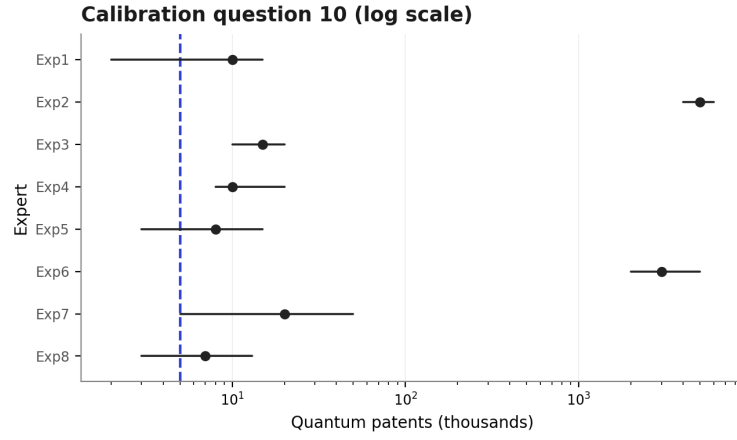


Figure 14: Range plot of calibration question 10.

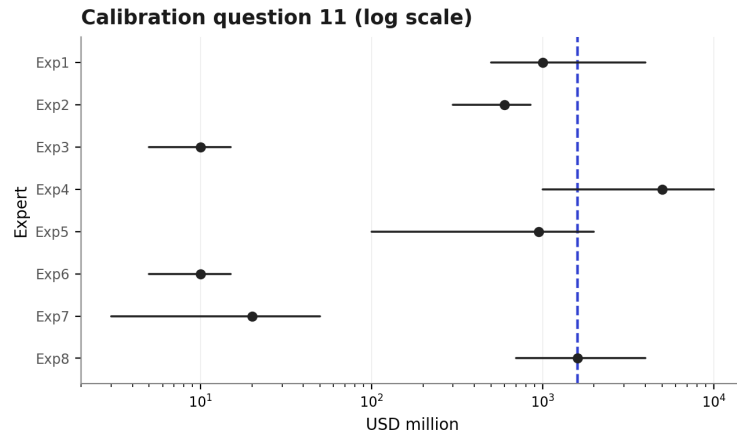


Figure 15: Range plot of calibration question 11.

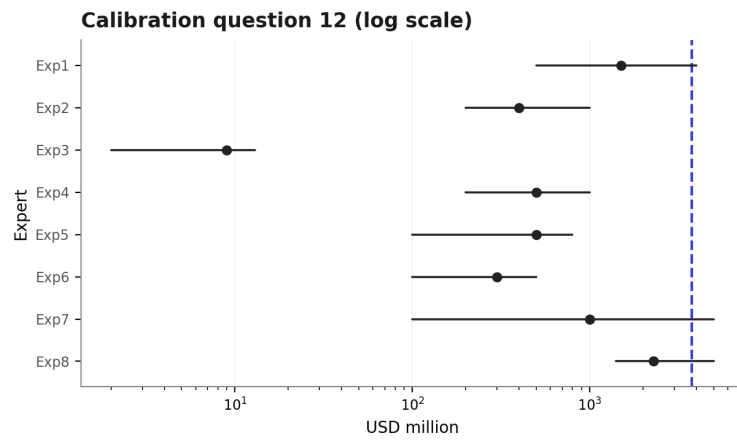


Figure 16: Range plot of calibration question 12.

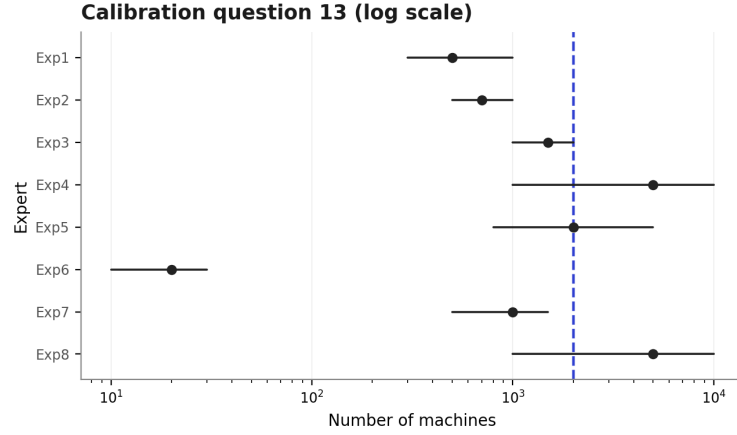


Figure 17: Range plot of calibration question 13.

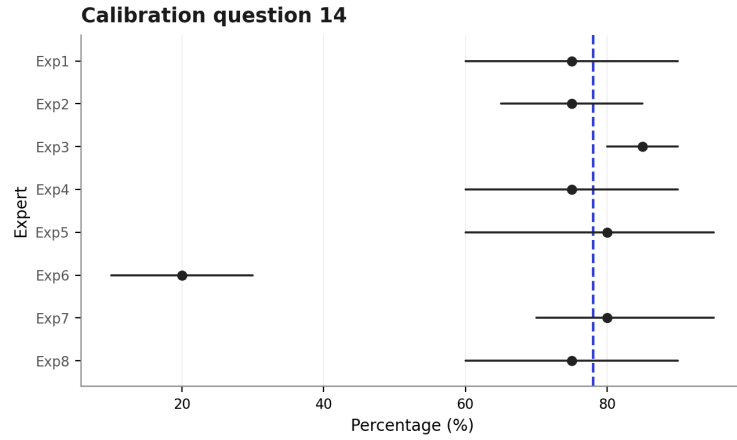


Figure 18: Range plot of calibration question 14.

C Questions of interest plots

For all experts the uncertainty quantification is visualized by means of using a line between the 5% and 95% quantile. Their 50% quantile estimation is visualized by a dot. Since the true realization does not exist, it is not plotted.

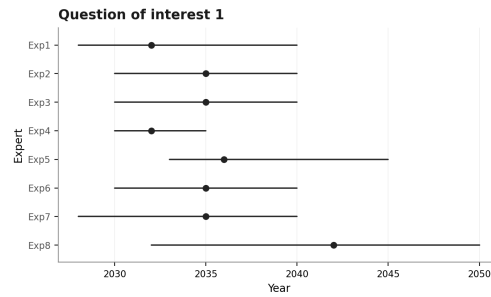


Figure 19: Range plot of question of interest 1.

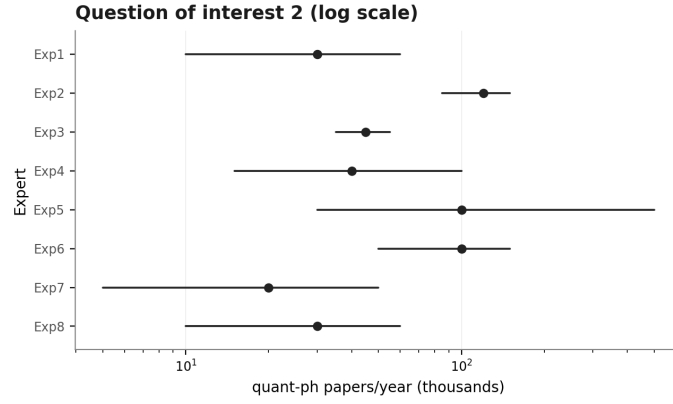


Figure 20: Range plot of question of interest 2.

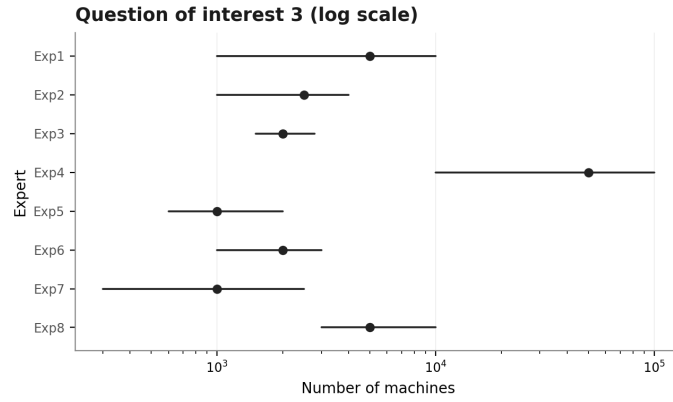


Figure 21: Range plot of question of interest 3.

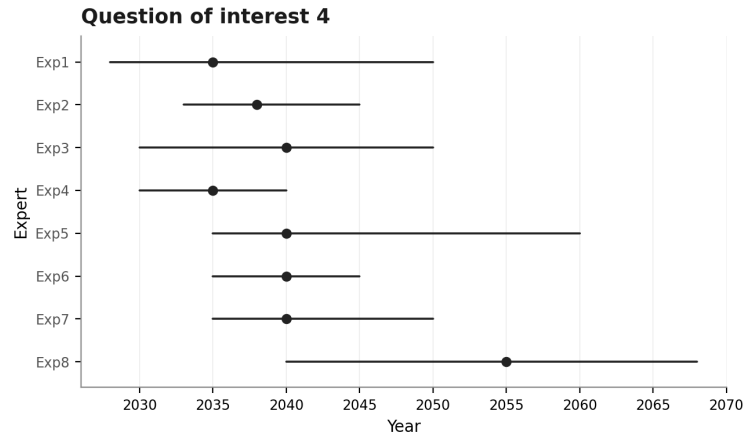


Figure 22: Range plot of question of interest 4.

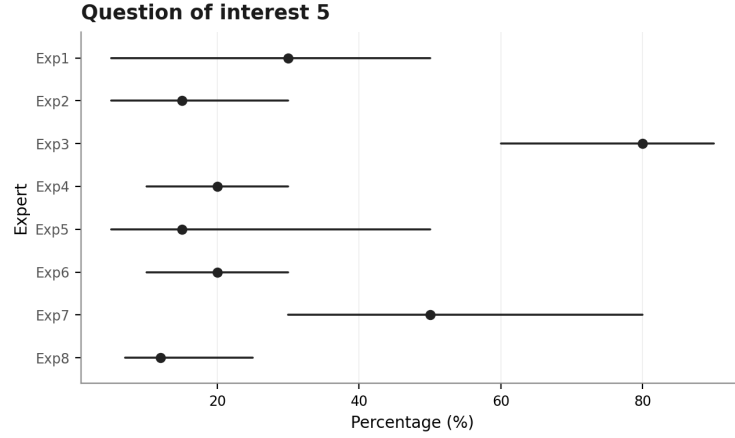


Figure 23: Range plot of question of interest 5.

D Questionnaire

The elicitation was conducted through a Google Forms questionnaire consisting of four sections. The first two sections provided background information and explained the elicitation process to the participants. The final two sections contained the calibration questions and the questions of interest. The link can be found here: <https://docs.google.com/forms/d/e/1FAIpQLScwQsk7d8Fyi2hI8EAM0kLgDfWpfQysfkR9w/viewform?usp=dialog>.